The Impact of Scalar Forcing on Cloud Microphysics Based on Direct Numerical Simulations

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Direct numerical simulations of turbulence-cloud-aerosol interactions have gained traction recently. The models are based on homogeneous and isotropic turbulence, the Eulerian transport of scalar quantities and the Lagrangian tracking of cloud droplets and aerosol particles. It is well known that turbulence decays unless a production mechanism exists. Although momentum forcing is common, scalar forcing has not received enough attention. In this work, we investigate two scalar forcing mechanisms, in the spectral and physical space, respectively. The scalars of interest in the present study are temperature, water vapor mixing ratio, and supersaturation. The statistics of forced scalar fields such as the probability density functions, and the spectra are analyzed. The impacts of scalar forcing on cloud and aerosol microphysics are also investigated. Our results show that forcing leads to more fluctuations in the scalar fields and broadens their probability distributions. At the same time, the rate of evaporation decreases. The scalar spectra for forced and unforced cases are different at small scales. The microphysics regime does not change from fast to slow when scalar forcing is applied.

I. Nomenclature

c_p, c_d	=	specific heat of air [J/(kg.K)], rate of condensation or evaporation [1/s]
ϵ, η	=	turbulent kinetic energy dissipation rate [m^2/s^3], Kolmogorov length scale [m]
$\boldsymbol{F}_b$, $\boldsymbol{F}_c$, $\boldsymbol{f}_c$	=	buoyancy force [N], external force [N], external force in the Fourier space [N]
F_T, F_{q_v}	=	temperature forcing term [K/s], vapor forcing term [g/kgs]
κ, L_h	=	hygroscopicity of solute [-], specific latent heat [J/kg]
μ_T, μ_v	=	thermal diffusivity in air [m^2/s], molecular diffusivity in air [m^2/s]
ν, p	=	kinematic viscosity of air [m^2/s], instantaneous fluid pressure [N/m^2]
q_c, q_v	=	instantaneous liquid water mixing ratio [g/kg], instantaneous vapor mixing ratio [g/kg]
r_d, r	=	dry aerosol radius [μm], wet radius of aerosol particle or cloud droplet [μm]
R, ρ_a, ρ_l	=	universal gas constant [J/(kg.K)], density of air [kg/m^3], density of water [kg/m^3]
S_e, S_k	=	environmental supersaturation [-], particle equilibrium supersaturation [-]
T, τ_p	=	instantaneous temperature [K], particle response time [s]
$\hat{\boldsymbol{u}}$, $\boldsymbol{V}$, $\boldsymbol{X}$	=	Fourier velocity [m/s], particle velocity vector [m/s], particle position vector [m]

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II. Introduction

Turbulence-cloud-aerosol interactions in the atmosphere are not properly understood and such knowledge gaps contribute to the uncertainties in the cloud-resolving climate models. Since experimental measurements are very few, numerical modeling can be a good alternative to produce required data for understanding the physics of the problem. The most accurate computational approach combines direct numerical simulation (DNS) of atmospheric turbulence with Lagrangian tracking of cloud and aerosol particles. In addition, two scalar quantities, namely temperature and water vapor mixing ratio, are transported in the Eulerian frame. The evolution of scalar fields directly influences the growth of clouds and aerosols. But fluctuations in the scalar fields decay with time. This decay might be prevented if scalar forcing is in effect. Therefore, the impact of forcing on the scalar fields and cloud microphysics in a DNS study demands more attention.

Vaillancourt et al. [1] coupled DNS with Lagrangian particle tracking and differentiated between the macroscopic and microscopic fields. The baseline solutions, which correspond to the constant density and hydrostatic conditions in the atmosphere, were referred to as the macroscopic field. The microscopic field consisted of the perturbations in the base solutions. Kumar et al. [2] studied turbulent mixing and entrainment at the cloud boundary using a DNS model that couples the Eulerian description of turbulent velocity, temperature, and water vapor mixing ratio fields with a Lagrangian description of cloud droplets that grow and shrink by condensation and evaporation, respectively. Lanotte et al. [3] combined the Eulerian method for turbulent velocity field with the Lagrangian method for cloud droplets. They discussed the role of turbulence on droplet condensation at moderate to high Reynolds numbers. Gao et al. [4] developed a direct numerical simulation (DNS) model to simulate entrainment and mixing for three initial configurations.

Unless a turbulence production mechanism exists, or an external forcing is applied, turbulence decays in numerical models. Therefore, forcing is required to sustain turbulence. This can be done in the physical and spectral or Fourier space. Eswaran and Pope [5] demonstrated that spectral forcing makes the turbulence statistically stationary. Rogallo et al. [6] devised an initialization procedure for homogeneous and isotropic turbulence (HIT). Janin et al. [7] applied a linear forcing scheme to maintain isotropic turbulence while controlling the integral length scale. Fourier forcing is applied to a band of low wavenumbers in the spectral space, whereas linear forcing is carried out in the physical space with energy being injected at all wavenumbers. Kumar et al. [2], Gao et al. [4], and Yu Li et al. [8] implemented momentum forcing to maintain HIT in the context of atmospheric clouds. But their models did not include scalar forcing.

Forcing can also been applied to sustain turbulence in the scalar fields (i.e. temperature and water vapor mixing ratio). Paoli and Shariff [9] applied spectral forcing in the temperature and vapor mixing ratio fields in addition to the momentum field to study cloud growth. Chen et al. [10] applied low wavenumber forcing to the two scalar fields to study mixed-phase clouds. Overholt and Pope [11] proposed a forcing scheme based on scalar mean gradients and showed that scalar fields reach a statistically stationary state. Carroll et al. [12] adopted a linear forcing method and examined it for a range of Schmidt numbers. Daniel et al. [13] constructed a scalar forcing term in the physical space based on reaction analogy and showed that their method maintains the scalar fields within predefined bounds.

Turbulence in a scalar field is crucial, especially at small scales. Yeung [14] considered the Lagrangian description of turbulence, with special emphasis on the transport of passive scalars subject to molecular mixing. The author reported that scalar dissipation is highly intermittent, and their autocorrelations become de-correlated in time more rapidly than turbulent kinetic energy (TKE) dissipation. Brethouwer et al. [15] presented a numerical analysis of the relations between the small-scale velocity and scalar fields in fully developed isotropic turbulence. They employed momentum forcing spectrally at low wavenumbers and applied mean scalar gradient forcing in the physical space. Warhaft [16] discussed the behavior of passive scalars during turbulent mixing and concluded that scalar intermittency occurs in the absence of velocity intermittency (i.e. the velocity fields can be strictly Gaussian) and exhibits anisotropy at small scales.

The impacts of scalar forcing on the evolution of scalar fields and particle growth are investigated in this paper assuming HIT. Momentum forcing is done in a low wavenumber band in the spectral space. Two scalar forcing schemes, Fourier forcing and mean gradient forcing, are implemented in the spectral and physical space, respectively. We also vary turbulence intensity (TKE dissipation rate) in the velocity field. The mean and fluctuating scalar fields evolve from nonzero initial conditions. By comparing unforced and forced scalar fields, we analyze scalar fluctuations at the large, intermediate, and small scales. Also, the role of scalar forcing on cloud droplet evaporation is discussed.

III. Governing Equations

Turbulent velocity field

The flow of turbulent air is assumed to be governed by the physical-space Navier-Stokes equations with the Boussinesq approximation: [17]

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}.\nabla)\boldsymbol{u} = -\frac{1}{\rho_a}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{F}_b + \boldsymbol{F}_c\,, \tag{1}$$

$$\nabla.\boldsymbol{u} = 0\,. \tag{2}$$

Here, $\boldsymbol{u}$ is the instantaneous air velocity vector, p is the instantaneous air pressure, ρ_a is air density, and ν is the kinematic viscosity of air. Two external forces are present in the right-hand side of Eq. (1). $\boldsymbol{F}_b$ is the buoyancy force defined as: [17]

$$\boldsymbol{F}_b = -\boldsymbol{g}\left[\frac{T - T_o}{T_o} + 0.608(q_v - q_{vo}) - q_c\right], \tag{3}$$

Above, $\boldsymbol{g}$ is the vector of the acceleration due to gravity, T is the instantaneous temperature, and q_v and q_c are respectively the instantaneous water vapor mixing ratio and liquid water mixing ratio. The subscript '0' denotes a reference value. Note that $\boldsymbol{F}_b$ couples the turbulent velocity field with the evolution of cloud droplets that occur at the microscale. $\boldsymbol{F}_c$ is the momentum forcing term required to maintain turbulence in the velocity field. [4]

Scalar transport equations

The transport equations for the instantaneous temperature T and water vapor mixing ratio q_v are written as: [10]

$$\frac{\partial T}{\partial t} + (\boldsymbol{u}.\nabla)T = \frac{L_h}{c_p}C_d + \mu_T\nabla^2 T + F_T\,, \tag{4}$$

$$\frac{\partial q_v}{\partial t} + (\boldsymbol{u}.\nabla)q_v = -C_d + \mu_v\nabla^2 q_v + F_{q_v}\,. \tag{5}$$

where L_h is the latent heat of condensation or evaporation as the case may be, c_p is the specific heat of air at constant pressure, μ_T and μ_v are thermal diffusivity and water vapor diffusivity in air, respectively. The term C_d is the rate of mass exchange between liquid water and water vapor and is defined below in Eq. (17). F_T and F_{q_v} are the forcing functions in the temperature and vapor mixing ratio fields, respectively, and are defined in Eqs. (20) and (21).

Growth of cloud and aerosol particles

In the atmosphere, aerosol particles grow and activate into cloud droplets in a supersaturated environment, while cloud droplets evaporate and deactivate into aerosol particles in a subsaturated environment. We use the word "particle" in a general sense to imply aerosol particle (particle radius is less than critical radius), or cloud droplet (particle radius is greater than critical radius). The particle equilibrium supersaturation S_k can be found as [18]

$$S_k = \frac{A}{r} - \kappa\frac{r_d^3}{r^3}\,. \tag{6}$$

where r_d is the dry aerosol radius, r is the wet radius of aqueous aerosol or cloud droplet, κ is the hygroscopicity parameter of the solute (dry aerosol), and A is the parameter: [18]

$$A = \frac{2\sigma_l M_l}{RT\rho_l}\,. \tag{7}$$

Above, (M_l, ρ_l, and σ_l) are respectively the molar mass, density, and surface tension of water; and R is the universal gas constant. The expression for critical radius (r_c) is [19]

$$r_c = \sqrt{\frac{3\kappa r_d^3}{A}}\,. \tag{8}$$

For an aerosol particle to be activated into a cloud droplet, it needs to grow beyond its critical radius. Similarly, for a cloud droplet to be deactivated into an aerosol particle, it needs to shrink below the critical radius.

The motion of each cloud droplet is represented by the Lagrangian description, which consists of the kinematics of the particle velocity and the Newton's law. The equations for droplet position and droplet velocity can thus be written as: [4]

$$\frac{d\boldsymbol{X}_i(t)}{dt} = \boldsymbol{V}_i(t)\,, \tag{9}$$

$$\frac{d\boldsymbol{V}_i(t)}{dt} = \frac{1}{(\tau_p)_i}[\boldsymbol{u}_i(\boldsymbol{X},t) - \boldsymbol{V}_i(t)] + \boldsymbol{g}\,. \tag{10}$$

Above, subscript 'i' denotes an individual cloud droplet, $\boldsymbol{V}$ is the droplet velocity vector, and $\boldsymbol{X}$ is the droplet position vector. The particle response time is τ_p, which can be written as [4]

$$(\tau_p)_i = \frac{2\rho_l r_i^2}{9\rho_a \nu}. \tag{11}$$

where ρ_l is the density of water and r_i is the radius of the i-th droplet. Note that Eq. (11) is valid for Stokes particles when the Reynolds number based on the relative velocity between particle and bulk flow is significantly less than unity. $\boldsymbol{u}_i$ is the air velocity at the i-th particle position and is obtained through a trilinear interpolation of the Eulerian field. Also, $\boldsymbol{g}$ is neglected when sedimentation (falling of cloud droplets relative to air) is neglected. The change in particle radius during condensation (or evaporation) can be written as [20]

$$\frac{dr_i(t)}{dt} = \frac{G}{r_i(t)}\left(S_e(\boldsymbol{X},t) - S_k(\boldsymbol{X},t)\right)_i. \tag{12}$$

where S_e is the environmental supersaturation from the Eulerian field and then interpolated to the particle position.

$$S_e = \frac{q_v}{q_{v,s}} - 1\,. \tag{13}$$

The $q_{v,s}$, which is the saturation water vapor mixing ratio, can be calculated from: [21]

$$q_{v,s} = 621.97\frac{p_{sat}}{p - p_{sat}}, \tag{14}$$

where p_{sat} is the saturation water vapor pressure, which is obtained from: [22]

$$p_{sat} = 611.2 \times \exp\left(\frac{17.67(T - 273.15)}{T - 29.65}\right), \tag{15}$$

In Equation (9), G is a growth parameter and can be found as: [20]

$$G = \frac{1}{\frac{L_h\rho_l}{k_T' T}\left(\frac{L_h}{R_v T} - 1\right)(1 + S_k) + \frac{\rho_l R_v T}{\mu_v' p_{sat}}}, \tag{16}$$

where R_v is the specific gas constant for water vapor. Also, k_T' and μ_v' are the modified thermal conductivity and modified water vapor diffusivity in air [20]. With the foregoing, the rate of mass exchange C_d is determined as: [23]

$$C_d(\boldsymbol{X},t) = \frac{4\pi\rho_l G}{\rho_a a^3}\sum_{i=1}^{n} r_i(t)\left(S_e(\boldsymbol{X},t) - S_k(\boldsymbol{X},t)\right)_i, \tag{17}$$

where m_a is the mass of air in each grid cell that has a volume of a^3. q_c is found as [4]

$$q_c(\boldsymbol{X},t) = \frac{4\pi\rho_l}{3\rho_a a^3}\sum_{i=1}^{n} r_i^3(t)\,. \tag{18}$$

Forcing mechanisms

The velocity field is initialized according to the method of Rogallo et al. [6] for homogeneous, isotropic turbulence. Turbulence will decay unless an external forcing is applied to maintain it. Although the gravitational body force term $\boldsymbol{F}_b$ could in principle maintain turbulence in the flow, the spatial dimensions involved in microscale physics is too small to permit a sustaining turbulence field. Thus, without production from shear, Dirichlet boundaries, or by baroclinic means, there is no natural mechanism available in our model to prevent a complete decay of the imposed initial turbulence field. The external force can be formulated in the spectral space as: [24]

$$\boldsymbol{f}_c(\boldsymbol{k},t) = \epsilon_{in}\frac{\widehat{\boldsymbol{u}}(\boldsymbol{k},t)}{\sum_{k_f \epsilon_k}\left|\widehat{\boldsymbol{u}}(\boldsymbol{k}_f,t)\right|^2}\delta_{k,k_f}\,. \tag{19}$$

Above, $\widehat{\boldsymbol{u}}(\boldsymbol{k},t)$ is the Fourier-transformed velocity field, $\boldsymbol{k}_f$ is chosen from a given wavenumber band, and δ_{k,k_f} is a delta function. Also, ϵ_{in} is the initial TKE dissipation rate. Turbulence intensity can be varied by changing the band of forced wavenumbers as well as by varying ϵ_{in}. However, our experience in this study has shown that the former is more strongly correlated with $\boldsymbol{f}_c$ than the latter. Hence, we choose to apply different wavenumber bands. Specifically, we have used the values of $1 \le k_f/k_{min} \le 3.9$ and $1 \le k_f/k_{min} \le 12.5$, where $k_{min} = 2\pi/L = 12.27$ is the minimum wavenumber and L is the domain length (0.512 m).

A forcing mechanism is required to sustain scalar turbulence. We are experimenting with two scalar forcing methods. The first is spectral forcing, applied to a band of low wavenumbers in the spectral space. Analogous to Eq. (19), we can write:

$$f_\theta(\boldsymbol{k},t) = \epsilon_{\theta,in} \frac{\hat{\theta}(\boldsymbol{k},t)}{\sum_{k_f \epsilon_k} \left|\hat{\theta}(\boldsymbol{k}_f,t)\right|^2} \delta_{k,k_f} \,. \tag{20}$$

Above, $\hat{\theta}(\boldsymbol{k},t)$ is the Fourier-transformed scalar field. The parameter $\epsilon_{\theta,in}$ is the initial scalar dissipation rate. Its unit is K^2/s for temperature and $g^2/kg^2 s$ for water vapor mixing ratio.
Note that $\boldsymbol{f}_\theta$ must be converted back to the physical space. The forcing wavenumber band is $1 \le k_f/k_{min} \le 3.9$.

The second method is mean scalar gradient forcing defined in the physical space as follows: [11]

$$F_{\theta,MG} = w\beta \tag{21}$$

where w is the instantaneous vertical velocity in the z-direction and β is the mean scalar gradient. The unit of β is for K/m for temperature forcing and g/kgm for vapor forcing in Eq. (21). We find the value of β as follows:

$$\beta = \langle \frac{f_\theta^{-1}(\boldsymbol{k},t)}{w} \rangle \tag{22}$$

IV. Numerical Implementation

We have used the FronTier CFD package [25] to carry out the calculations. A fifth order finite-difference based Weighted Essentially Non-Oscillatory (WENO) scheme is used to solve the convective terms in Eqs. (1), (4) and (5). The diffusion terms in those equations are solved using the Crank-Nicolson scheme. The PETSc package [26] is adopted to iteratively solve the systems of linear equations arising from the discretization of the diffusion terms. A projection scheme is used, and the Poisson equation is solved for the pressure field. The equations in (9) and (10) are solved using the Implicit Euler method. The MPI protocol is utilized for parallelization.

V. Computational Setup

Table 1 lists the statistics of turbulent velocity field such as the TKE, its dissipation rate, ϵ, the Taylor microscale, λ, the Kolmogorov length scale, η, the integral length scale, l_0, the forcing Reynolds number, Re_f, the Taylor-scale Reynolds number, Re_λ, and the Reynolds number based on the integral scale, Re_{l_0}.

Table 1: Statistics of turbulent velocity fields at 4 s

Turbulence level	TKE (m^2/s^2)	ϵ (m^2/s^3)	$\lambda\ (m)$	$\eta\ (m)$	$l_0\ (m)$	Re_f	Re_λ	Re_{l_0}
High	0.0094	0.00494	0.0169	0.00091	0.184	401.12	156.11	1704.7
Low	0.0025	0.00055	0.0261	0.00157	0.227	192.97	138.74	1207.6

The computational domain is set to be $0.512^3\ m^3$. The domain is divided into subsaturated clear-air region [4] and supersaturated cloudy region. The boundary conditions are triply periodic. The initial cloud fraction is 0.5 and a slab-like cloud configuration is assumed. The initial vapor mixing ratio field is specified as [4]

$$q_v(z,t_0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \le z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & elsewhere \end{cases} \,. \tag{23}$$

Above, $q_{v,s} = 3.873\ g/kg$, $q_v^{max} = 1.02 \times q_{v,s} = 3.949\ g/kg$, and $q_{v,e} = 0.30\ g/kg$. Also, $d = L/2$ is the width of the cloud slab. The temperature field is initialized as [4]

$$T(t=0) = T_o - 0.608 T_o (q_v(t=0) - q_{vo}) \,. \tag{24}$$

where, T_o and q_{vo} are the domain averages.

Table 2: Test cases

Case	Scalar forcing type	Turbulence intensity	Initial droplet size	Dry aerosol size (μm)	Grid resolution
L	N/A	Low	10 μm	0.126 ± 0.02	0.002 m
L-F	Fourier	Low	10 μm	0.126 ± 0.02	0.002 m
L-MGF	Mean gradient	Low	10 μm	0.126 ± 0.02	0.002 m
H-F	Fourier	High	10 μm	0.126 ± 0.02	0.002 m

At the beginning, a total of 16×10^6 cloud droplets, each having a radius of 10 μm, are randomly placed in the cloudy region. We simulate four cases (listed in Table 2). The dry aerosol size distribution is lognormal with a mean of 0.126 μm and standard deviation of 0.02 μm. The value of κ is 0.61 for ammonium sulfate composition. The identifiers 'L' and 'H' denote 'low' and 'high' turbulence levels, respectively. The identifiers 'F' and 'MGF' indicate 'Fourier' and 'mean gradient' scalar forcings, respectively. Note that momentum forcing is implemented in the Fourier space for all the four cases. The grid resolution is 256^3 and the $k_{max}\eta$ values are 2.46 and 1.43 for 'low' and 'high' turbulence levels, respectively. The Bachelor length scales ($\eta_B = \eta/\sqrt{\nu/\mu_T}$) are 0.00188 m and 0.00109 m for 'low' and 'high' turbulence levels, respectively.

VI. Results and Discussion

We present our results in this section.

Forced and unforced scalar fields

The standard deviation of a scalar field, σ_θ can be found as

$$\sigma_\theta = \sqrt{\frac{\sum_i(\theta_i - \langle\theta\rangle)^2}{N}}, \tag{25}$$

where θ is a placeholder for scalar quantity and N is the number of grid points. Figure 1(a) shows the probability density functions (PDF) of the fluctuating temperature, $T' = T - \langle T\rangle$, normalized by the standard deviation, σ_T. Figure 1(b) is the corresponding result for water vapor mixing ratio. We see that forcing increases the amplitudes of the fluctuations in the scalar fields as the PDFs are broader for Cases L-F and L-MGF compared to the unforced case (L).

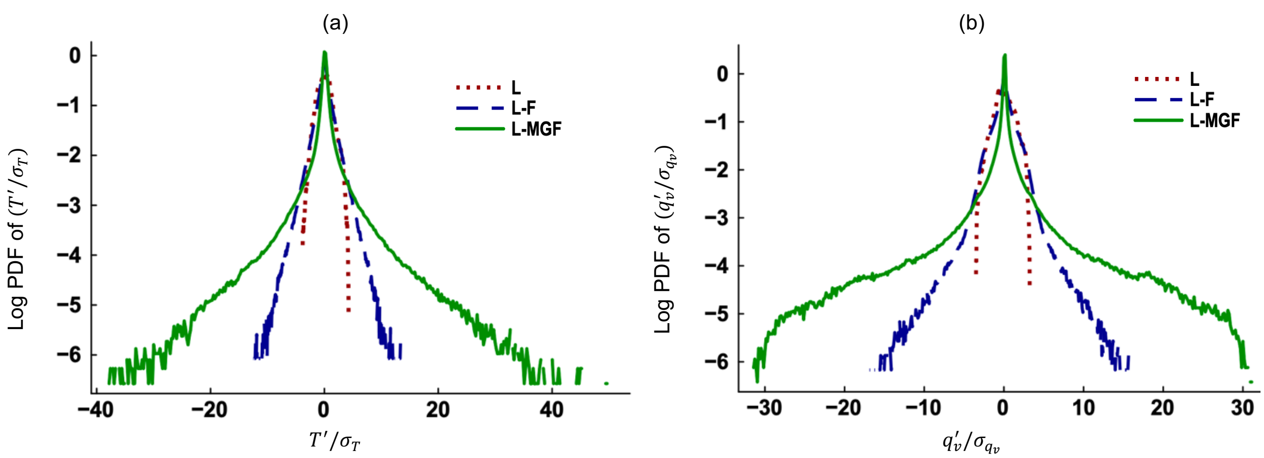

Fig. 1 Log PDFs of (a) T'/σ_T and (b) q_v'/σ_{q_v}, for the three cases at 25 s.

With mean gradient forcing, the amplitudes of fluctuations are larger compared to those for Fourier forcing. Note that the mean gradient in our study is not a constant unlike in Overholt and Pope [11]. Instead, we update the mean gradient at every timestep based on Eq. (22). The presence of two scalar transport equations with source (latent heat) terms increase the complexity of our problem. The latent heat describes the phase change events like condensation or evaporation and couples the scalar fields with particles.

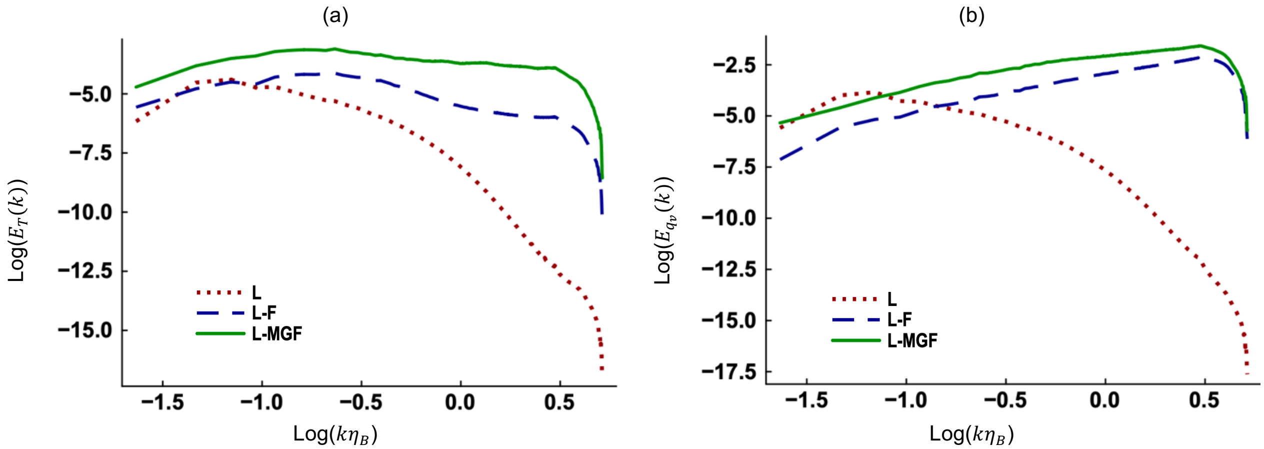

Fig. 2 Spectra of (a) temperature and (b) vapor mixing ratio, at 20 s.

The spectra of temperature and vapor mixing ratio are plotted in Figs. 2(a)-(b) for Cases L, L-F, and L-MGF at 20 s. We find that these spectra differ at all scales, particularly at small scales. Kolmogorov's theory suggests that small-scale structures should be statistically homogeneous and isotropic. [27] But the scalar fields are not homogeneous and isotropic at small scales unlike the velocity field. The scalar fields are coupled with particles via the latent heat (C_d) in Eqs. (4) and (5). Again, particles are smaller than the Kolmogorov length scale. This makes the cascading process in the scalar fields different from the velocity field. Although, the buoyancy force is present in Eq. (1), it is negligible in this study due to the small domain size (0.512 m) relative to the real atmosphere. We also point out that there is a loss of accuracy in calculating the C_d. As particle size is smaller than the grid spacing, interpolation to the Eulerian grid points is required. The mean scalar fields in this study are much larger than their fluctuations. Hence, the former might influence the latter when the evolution of the total field (mean plus fluctuating) is computed.

Scalar forcing and cloud microphysics

Figure 3 shows the time evolution of mean radii of all particles (cloud droplets plus aerosol particles). Without scalar forcing, the mean radius decreases the most. Increasing scalar fluctuations (Case L-MGF in Fig. 1) slow down the rate of evaporation (Fig. 3). If the flow turbulence level increases (Case H-F), the evaporation rate is further reduced (Fig. 3). Note that scalar fluctuations are smaller than the mean values by orders of magnitude and that scalar forcing significantly affects the mean radii of particles. It appears that particle growth reached a steady state at different levels depending on the levels of scalar and flow turbulence. Also, more turbulence leads to equilibrium faster because of stronger mixing. This finding has atmospheric implications. The strength of mixing and entrainment varies in different regions of the cloud. [28] By changing the magnitudes of forcing, diverse regions in the cloud can be simulated.

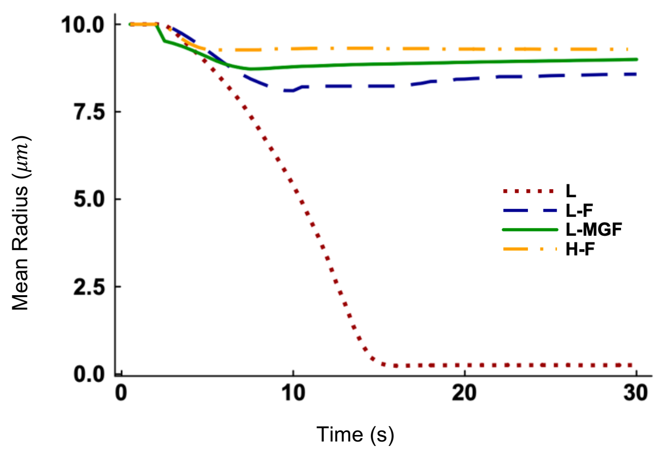

Fig. 3 Time evolution of the mean radii of all particles for Cases L, L-F, L-MGF, and H-F.

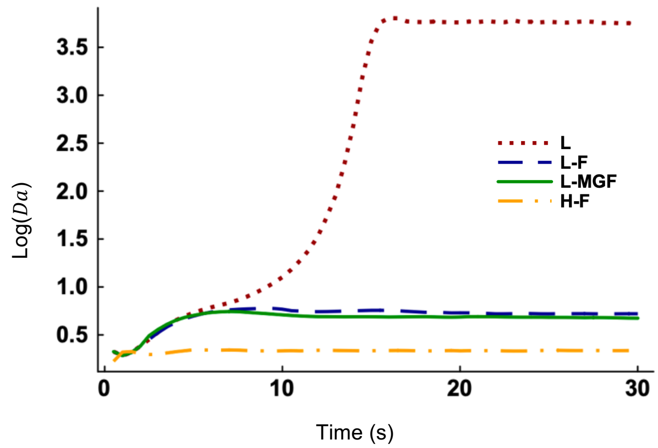

Fig. 4 Time evolution of the Damkohler number for Cases L, L-F, L-MGF, and H-F.

The Damkohler number (Da) characterizes the relationship between turbulent mixing and microphysics. It is defined as the ratio of turbulent mixing timescale (τ_{mix}) and droplet evaporation timescale (τ_{evap}) [29]. Figure 4 plots the time evolution of Da on a logarithmic scale. We see that $\tau_{mix} < \tau_{evap}$ at 0 s. For the unforced case (L), τ_{evap} becomes smaller than τ_{mix} by 10 s as scalar fluctuations decay in the absence of forcing. On the other hand, the microphysics regime remains fast ($\tau_{evap} > \tau_{mix}$) throughout since fluctuations in the scalar fields are sustained by forcing.

VII. Conclusion

We have investigated turbulence-cloud-aerosol interactions using a DNS model that incorporates scalar forcing. Such models can simulate microphysical events in different regions of the cloud by sustaining scalar turbulence.

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